

Problem 18.18

A perpetuity paying 1 at the beginning of each 6-month period has a present value of 20. A second perpetuity pays X at the beginning of every 2 years. Assuming the same annual effective interest rate, the two present values are equal. Determine X .

Solution

Let j be the effective interest rate per 6-month period. Since the first perpetuity is paid at the beginning of each 6-month period, it is a perpetuity due. Thus,

$$20 = \ddot{a}_{\infty|j} = 1 + \frac{1}{j}.$$

Therefore,

$$\frac{1}{j} = 19 \quad \implies \quad j = \frac{1}{19}.$$

The annual effective interest rate i satisfies

$$1 + i = (1 + j)^2 = \left(1 + \frac{1}{19}\right)^2,$$

so

$$i = \left(\frac{20}{19}\right)^2 - 1.$$

Let k be the effective interest rate over a 2-year period. Since 2 years is two annual periods,

$$1 + k = (1 + i)^2.$$

Using $1 + i = (20/19)^2$,

$$1 + k = \left(\frac{20}{19}\right)^4,$$

and hence

$$k = \left(\frac{20}{19}\right)^4 - 1 = \frac{29679}{130321}.$$

The second perpetuity is also a perpetuity due, with payments every 2 years. Therefore,

$$20 = X\ddot{a}_{\infty|k} = X\left(1 + \frac{1}{k}\right).$$

Solving for X ,

$$X = \frac{20}{1 + 1/k} = 20\frac{k}{1 + k}.$$

Substitute $k = 29679/130321$:

$$X = 20 \frac{29679}{160000} = 3.709875.$$

Thus,

$$\boxed{X = 3.709875}.$$